



2026-27

CONCEPTS | CLINICALS | MCQS

PSYCHIATRY

- Understand Minds. Answer Right.
- Classification, drugs & key clinical features.
- High-yield case scenarios simplified.

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PSYCHIATRY

DISTURBANCES OF THOUGHT

1. Flow/Stream of thought-

a. Flight of ideas- rapid, superficially connected shifting of ideas, typical of mania.

Eg- "I need to go to the market. Markets are crowded. Crowds cheer in stadiums. Stadiums have cricket matches. I love cricket. Cricket bats are made of wood."

b. Perseveration- Repetition of the same response beyond the point of relevance.

Eg- Examiner: "What is your name?" → Patient: "John."

Examiner: "Where do you live?" → Patient: "John."

Examiner: "What is your age?" → Patient: "John."

c. Circumstantiality- Circle around → but finally reach the point

FMGE 2025

Eg- Question: "Do you work?"

Patient:

"Yes, I do. You know, finding a job these days is really difficult. When I finished college, I wasn't sure what to do. My parents wanted me to try engineering, but I was more interested in teaching. Anyway, after some struggle, I got a teaching job at a school."

d. Thought block- seen in severe anxiety

2. FORMAL THOUGHT DISORDER

NEETPG 2021

a. Loosening of association- Loss of connection between components of the same thought.

eg-Examiner: "Do you have pain?"

Patient: "Pain is a window. Windows let air. Air is important for life. Life is a movie. Movies are made in Bombay."

b. Derailment- lack of association between two successive thoughts

eg-Examiner: "Do you like going to the market?"

Patient: "Yes, the market has many shops. Shops sell books. I like reading. Reading is done at night."

c. Incoherence- word salad

d. Tangentiality- roundabout, never reaches the point.

eg-Examiner: "Do you have children?"

Patient: "Children are the future of the nation. If we take care of them, the country will grow stronger. Schools must be better funded."

e. Neologism- forming new words which does not have any meaning

3. CONTENT OF THOUGHT

NEETPG 25, FMGE 21, 20, FMGE 2026

a. Delusion of persecution- most common type of persecution → someone wants to kill me/harm me.

eg-“My neighbors have installed cameras in my house walls to watch me.”

b. Delusion of reference- The patient believes that neutral events happening around him are somehow related to him.

eg-“Those people laughing over there are talking and laughing about me.”

c. Delusion of grandiosity- patient feels that he has super powers.

d. Delusion of love- de -Clarambault syndrome/erotomania

FMGE 21

eg-A patient believes a famous actor is secretly in love with them and sends hidden love messages through movies or interviews.

NEET PG 2020, INI CET 23

e. Delusion of negation-Nihilistic delusion/Cotard's syndrome

eg- a patient with severe depression claimed that all his internal organs have rotted.

Fmge 2026

F.) delusion of control: False belief that a person's will, thoughts, or feelings are being controlled by external forces.

g) Delusion of infidelity- morbid jealousy/pathological jealousy/Othello syndrome

eg-Husband insists that his wife is having an affair with the milkman because she smiled when receiving the milk.

1. Capgras Syndrome = “Imposter”

Belief: Loved one replaced by a look-alike imposter.

Example: “This woman looks like my wife, but she’s not my wife.”

Trick: Capgras → Copycat (imposter copy).



2. Fregoli Syndrome = “Disguise”

Belief: Many people are actually one single persecutor in disguise.

Example: “My doctor, neighbor, and shopkeeper are all the same person—just changing disguises.”

Trick: Fregoli → Frequent roles (one person → many roles).



4. Disorders of Possession

- a. **Thought insertion**- My neighbor is putting thoughts in my mind.
- b. **Thought withdrawal**- My neighbor withdrew/stole thoughts from my mind.
- c. **Thought broadcast**- The patient experiences that thoughts are escaping his mind, and others can access them.
- d. **Obsessions**- : A thought comes repeatedly into the patient's mind against his will.

Assessment of Higher Mental Functions

FMGE 23

Digit Span Test- to check for attention, specifically, immediate memory

Serial Seven Subtraction Test- to check for concentration

INSIGHT STAGES

- I- insight absent, complete denial of illness
- II- slight awareness of being sick, but denying at the same time
- III- awareness of being but attributing to external factors
- IV- awareness of being sick but attributing to something unknown to self
- V- awareness of being ill attributes to oneself not accepting for treatment

SCHIZOPHRENIA

NEETPG 2023, INICET 25, FMGE 23, NEET 25, FMGE 26

4 A's of Bleuler

AUTISM - socially withdrawal, living alone

AMBIVALENCE- inability to take decision

AFFECT DISTURBANCES- anhedonia, mania, depression

ASSOCIATION DISTURBANCES- formal thought disorder

The first rank symptoms of schizophrenia described by Kurt

Schneider include:

1) Hallucination:

- * First-person hallucination
- * Second-person hallucination (voices directly addressing the patient)
- * Third-person hallucination (voices speaking about the patient)

2) Thought phenomenon:

- * Thought withdrawal
- * Thought insertion
- * Thought broadcasting

3) Made phenomenon:

- * Made affect
- * Made impulses or drives
- * Made volitional acts

4) Delusional perception : fixed and false belief

5) Somatic passivity

5. Somatic Passivity - The patient experiences somatic sensations & blames an external agency for the same.

Pathogenesis

Neurotransmitter hypothesis

Dopamine hypothesis	Excessive levels of dopamine cause schizophrenia
Dopamine and serotonin hypothesis	Excessive levels of dopamine & serotonin cause schizophrenia

SYMPTOMS

FMGE 2025

Positive symptoms- respond well to medications	Negative symptoms -Respond poorly to medications
Delusions - M/C is delusion of persecution Hallucinations - M/C - auditory hallucinations	Avolition - Loss of drive for goal-directed activities Apathy - Lack of concern Anhedonia - Lack of pleasure in previously pleasurable activities Asociality - Lack of social interaction Affective flattening (Or emotional blunting) - Lack of emotional response Alogia - Decreased verbal communication
Cause- Dopamine excess in the mesolimbic tract (ventral tegmental area to nucleus accumbens) leads to positive symptoms.	Cause- Decreased dopamine in the mesocortical tract (Ventral tegmental area to the prefrontal cortex) results in negative symptoms.

Motor symptoms/catatonic symptoms

Stupor- A state of extreme inactivity or immobility



Waxy flexibility- initial few resistance then take a new posture



Posturing- Maintenance of a posture for a long period of time.

Negativism

Catalepsy



Echolalia - Repetition of speech

FMGE 22

Echopraxia- Repetition of behaviour

Grimacing - Maintenance of odd facial expressions

Diagnosis-

DSM-V criteria- 1-6 months of duration → Schizophreniform illness

DSM-V criteria- >6 months of duration → Schizophrenia

ICD-11 criteria - <1 month of duration → Acute Transient Psychotic (ATP) disorder

ICD-11 criteria- >1 month of duration → Schizophrenia

TREATMENT

Antipsychotics are the mainstay of treatment in schizophrenia

(in catatonia, the first line treatment is i.v lorazepam and electroconvulsive therapy)

	Typical	Atypical
Mechanism	D2 antagonism	D2 and 5HT2 antagonism
Effective against	Positive symptoms	Positive and negative symptoms
Extrapyramidal symptoms & hyperprolactinemia	more	less

Metabolic side effects	Less	More
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FMGE 2025,INI CET 22

Typical antipsychotics - block D2 receptor only (First generation antipsychotics)	Atypical antipsychotics - block D2 receptor and 5HT receptor (Second generation antipsychotics)
• Phenothiazines- Chlorpromazine, thioridazine, prochlorperazine, fluphenazine • Thioxanthenes- Thiothixene, flupenthixol • Butyrophenones- Haloperidol, droperidol	• Clozapine, olanzapine • Risperidone, paliperidone, iloperidone • Quetiapine, ziprasidone, aripiprazole • amisulpride • Newer ones - Brexpiprazole, cariprazine, pimavanserin
Drug-induced parkinsonism Symptoms - Tremors, Rigidity, Bradykinesia Prophylaxis - Anticholinergics → trihexyphenidyl Treatment - Anticholinergics (E.g. trihexyphenidyl)	Side effects of atypical antipsychotics Metabolic side effects: More common than typical antipsychotics. (clozapine and olanzapine has highest metabolic side effects)
Acute dystonia FMGE 2024 Symptoms - Torticollis → twisted neck Oculogyric crisis → upward eye deviation Opisthotonus → arching of back Jaw spasm, tongue protrusion, laryngospasm Earliest side effect of antipsychotics Treatment - Parenteral anticholinergics (e.g. i.m. promethazine) Prophylaxis - Anticholinergics (E.g. trihexyphenidyl, diphenhydramine)	Clozapine Most effective antipsychotic Drug of choice in TRS Max weight gain property Only antipsychotic with anti suicidal property S/E- Agranulocytosis Myocarditis seizures (dose-dependent) Sedation (most common) Sialorrhea ANC and TLC monitoring is required Stop clozapine, if WBC < 3000/dl or ANC < 1500/dl
Acute akathisia Commonest side effect of antipsychotics Characterised by an inner sense of restlessness along with objective signs of restlessness DOC - Propranolol	
Tardive dyskinesia NEETPG 2025 Orofacial movements (most common) Lip smacking, chewing, tongue protrusion, grimacing Writhing of fingers, limbs Cause - Long term use of antipsychotics and accompanying blockade of D2 receptors results in D2 receptors up-regulation along with postsynaptic	Imp- FMGE 2024 antipsychotic with least weight gain → Ziprasidone, aripiprazole, amisulpride, asenapine

dopamine receptor supersensitivity. Treatment - Shift to second generation antipsychotics, Use of Valbenazine, Tetrabenazine	
Neuroleptic malignant syndrome Key Features (MCQ classic = FEVER) F – Fever (very high, > 40°C) E – Encephalopathy (confusion, delirium, stupor, coma) V – Vital instability (BP fluctuations, tachycardia) E – Elevated CPK (muscle breakdown, myoglobinuria, renal failure) R – Rigidity (lead-pipe rigidity hallmark) Treatment- dantrolene Keep the patient antipsychotic-free for 2 weeks, then start with second-generation antipsychotics.	
Endocrine side effects Blockade of dopamine receptors in the tuberoinfundibular tract causes hyperprolactinemia Symptoms- Galactorrhoea, menstrual disturbances in females, low libido in males	

Acute Psychotic Disorders

FMGE 20

symptom-Delusions, hallucinations

Fever is a common stressor

ICD 11- <1 month symptom→ acute and transient psychotic disorder

DMS V- < 1 month→ Brief psychotic disorder

1-6 months→ Schizophreniform disorder

Trt- antipsychotic/benzodiazepines

DEPRESSION

INI CET 25, FMGE 21, 20, FMGE 2026

characterized by persistent low mood, loss of interest/pleasure, and associated cognitive + physical symptoms, lasting ≥2 weeks

ETIOLOGY

Neurotransmitter Hypothesis

↓ Serotonin (5-HT) → low mood, suicidality

↓ Norepinephrine (NE) → ↓ energy, attention

↓ Dopamine (DA) → anhedonia, loss of motivation

Mnemonic for Symptoms: SIGECAPS

Sleep disturbance

Interest ↓
 Guilt/worthlessness
 Energy ↓
 Concentration ↓
 Appetite change
 Psychomotor changes
 Suicidal thoughts

SPECIFIERS

Psychotic symptoms/features (psychotic depression)	Atypical features (atypical depression)	Melancholic features (melancholic depression)
mood-congruent delusions/hallucinations	mood reactivity, ↑ appetite, hypersomnia	early morning awakening, weight loss, psychomotor retardation

PHYSICAL SIGNS

Omega sign	Veraguth fold
 Omega shaped fold at the root of nose	 Triangular folds in upper eyelid

BECK'S COGNITIVE TRIAD

1. Negative view of Self→“I am worthless.”
2. Negative view of the World→“Nobody loves me.”
3. Negative view of the Future→“Things will never get better.”

INI CET 2024

TRT-

Antidepressants

INICET 22,FMGE 23

TCA-tricyclic antidepressant	Selective serotonin reuptake inhibitors (SSRIs)	.Serotonin-norepinephrine reuptake inhibitors (SNRIs)
Block reuptake of NE & 5-HT → ↑ availability in synapse Also block muscarinic (M), H1, and α1 receptors → side effects Drugs- Imipramine, desipramine, Amitriptyline,	Selectively block 5-HT reuptake Drugs- Fluoxetine, Sertraline, Escitalopram, Paroxetine, Dapoxetine, Fluvoxamine	Block reuptake of 5-HT & NE (but no major H1/α1/M1 block) Venlafaxine, Desvenlafaxine, Duloxetine Side effects- Similar to ssri with ↑ BP and ↑ HR

nortriptyline, clomipramine Clomipramine- most serotonin-selective TCA • Side effects-BLOCK B → Blurred vision (anticholinergic) L → Low blood pressure (α_1 block → postural hypotension) O → Overweight (H1 block → weight gain, sedation) C → Constipation & urinary retention (anticholinergic) K → Cardiac toxicity (arrhythmias, QT prolongation, QRS widening) Trt- Antidote → IV Sodium Bicarbonate (for arrhythmia)	Side effects - GI upset (Nausea, diarrhea) - Sexual dysfunction - Insomnia / agitation - Serotonin syndrome (if combined with MAOI) Serotonin syndrome -Due to excessive serotonin synthesis, inhibition of metabolism, or excessive action on receptors Diarrhea Rash Autonomic stability Seizures Hyperreflexia Trt- Cyproheptadine	
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Atypical antidepressants

Mirtazapine-

Noradrenergic & Specific Serotonergic Antidepressant

Minimal sexual side effect

Bupropion

FMGE 23

Norepinephrine Dopamine Reuptake Inhibitor

Minimal risk of sexual side effects, weight gain or sedation

Trazodone

SARI – Serotonin Antagonist and Reuptake Inhibitor, side effects- priapism

IMPORTANT NOTE-

Pregnant and lactating female with mild depression- CBT is preferred

Antidepressant of choice in pregnancy-sertraline

ECT- depression with suicidal risk

repetitive Transcranial Magnetic Stimulation- major depressive disorder

FDA has approved a nasal spray of esketamine (s-enantiomer of ketamine) for treatment-resistant depression

FMGE 2024,21

Recurrent depressive disorder- History of more than one depressive episode.

Dysthymia (Dysthymic disorder)- Depressive symptoms for more than 2 years

POST PARTUM PSYCHIATRY DISORDER

NEETPG 2023,22,20

Postpartum blues	Postpartum psychosis	Postpartum depression
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2–3 days after delivery, peaks at day 5, resolves by 2 weeks	Within 2–3 weeks postpartum	Within 4–6 weeks postpartum
Symptoms- tearfulness, sadness, mood lability and sleep disturbances	Symptoms- Delusions (baby is evil, baby is dead, persecution) - Hallucinations - Severe mood disturbance (mania or depression) - Risk of infanticide & suicide	Symptoms- Sadness, tearfulness, lability, sleep disturbances, anhedonia, suicidal thoughts/thoughts of harming baby
Self limiting Reassure the mother	Antipsychotics ± mood stabilizers	SSRIs (first-line), psychotherapy

Exam Trick:

If mother is sad but caring for baby → Blues

If mother is sad + neglects baby → Depression

If mother wants to harm baby or hears voices → Psychosis

FMGE 2026

- **HELLER SYNDROME** (Childhood Disintegrative Disorder)
normal development for at least the first 2 years e, followed by a marked regression in multiple developmental domains. The key clinical feature loss of previously acquired bladder control (urinary incontinence) along with decline in language, communication, social Interaction, and cognitive abilities.

PERSONALITY DISORDER**NEETPG 24,25, FMGE 2026**

Cluster A – “Odd & Eccentric” (Schizo-type)	Cluster B – “Dramatic, Emotional, Erratic”	Cluster C – “Anxious, Fearful”
Paranoid PD → Distrust, suspicious, interprets others as malevolent. Schizoid PD → Prefers isolation, emotionally cold, no desire for relationships. Schizotypal PD → Odd beliefs, magical thinking, eccentric behavior.	Antisocial PD → Disregard for rights of others, deceitful, criminality, no remorse. Borderline PD → Instability in mood, relationships, self-image, impulsive, fear of abandonment, self-harm. Histrionic PD → Attention-seeking, dramatic, shallow, seductive. Narcissistic PD → Grandiosity, need admiration, lack empathy.	Avoidant PD → Social inhibition, hypersensitive to criticism, wants relationships but avoids. Dependent PD → Excessive need to be taken care of, submissive, fear of separation. Obsessive-Compulsive PD (OCPD) → Preoccupation with order, perfectionism, control (different from OCD → no insight, ego-syntonic).

Bipolar Disorder**FMGE 25,20,INI CET 22**

Type 1- Mania/Mixed + Depression

Type II- Depression + hypomania only

Mania is caused by a hyperdopaminergic state

NEET PG 2021, INI 2025,24

Mania	Hypomania	Cyclothymia	Rapid Cycling
≥ 1 week	≥ 4 days	symptoms ≥ 2 years	≥ 4 mood episodes/year
Symptoms- Flight of ideas overTalkitiveness Irritable mood	Symptoms are less severe than mania	-	-
First-line: Mood stabilizers + antipsychotics	mood stabilizers + Antidepressant	-	Rx : Divalproate (DOC)

Lithium**INI CET 23,22,FMGE 22**

Mood stabilizer

Rapidly absorbed after oral intake, Body handles lithium like Na⁺

Uses-

Bipolar disorder - acute mania and maintenance treatment (only mood stabiliser with antisuicidal property)

-Doesn't bind to plasma proteins, not metabolised, and is excreted unchanged through the kidney.

Lithium has a narrow therapeutic index.
For acute mania- 1.0-1.2 mEq/L
For maintenance treatment- 0.4-0.8 mEq/L
Lithium toxicity - > 1.5 mEq/L

Side Effects

Mnemonic = LITHIUM

L – Leukocytosis (benign)

I – Insipidus (Nephrogenic diabetes insipidus → polyuria, polydipsia)

T – Tremor, Teratogenic (Ebstein anomaly in 1st trimester)

H – Hypothyroidism, Hyperparathyroidism ($\uparrow$ Ca^{2+})

I – Increased weight

U – Unwanted GI (nausea, diarrhea)

M – Misc: acne, edema, cognitive dulling

Toxicity + confusion + ataxia → hemodialysis

PANIC DISORDER

NEETPG 2023,20,FMGE 21,20

It's an acute attack of intense anxiety, with a 'feeling of impending doom'

Patient feel 'he is having a heart attack', 'he is about to die'

Lasting in 5 - 15 minutes

symptoms-Palpitations, choking sensations, chest pain, dizziness, depersonalisation

Mx- SSRI/SNRI

GENERALISED ANXIETY DISORDER

NEETPG 2024,FMGE 20

Most common anxiety

More common in female

Symptom- >6month

Free floating anxiety

Motor tension/restlessness : Inability to relax

Trt- Benzodiazepines (short term, tapered and stopped) & SSRIs (long term).

CBT is psychotherapy used.

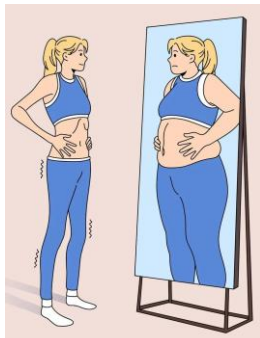
SPECIFIC PHOBIAS

FMGE 2024

Nyctophobia	Dark
Acrophobia	Heights
Claustrophobia	Closed space
Pyrophobia	Fire
Xenophobia	stranger

Generalised phobia- SSRI

Specific phobia- propranolol

OCD	BODY DYSMORPHIC DISORDER	HYPOCHONDRIASIS	HOARDING DISORDER
Intrusive obsessions (thoughts) and compulsions (acts/rituals)	Preoccupation with an imagined defect in appearance, or exaggerated concern about a minor physical flaw.	Preoccupation with having or developing a serious illness	Collecting items of no emotional/financial value
Fears (contamination, harm, symmetry, intrusive thoughts)	Appearance/looks (skin, nose, hair, body part)	Cancer, HIV, heart disease, etc.	Items/objects
Repetitive rituals (washing, checking, counting) to reduce anxiety	Repetitive checking of mirror, grooming, reassurance seeking	Frequent doctor visits, health checks, internet searches	Accumulation of items → cluttered living spaces
SSRIs (high dose) + CBT (ERP)	SSRIs + CBT (esp. ERP, cognitive restructuring)	CBT (illness-focused) +/- SSRIs	Trt- SSRI and CBT
Exposure and response prevention- Therapy of choice			
			

EXAM TRICKS-

OCD → Thoughts + rituals (ego-dystonic).

BDD → Appearance-focused, mirror checking.

Illness Anxiety → Fear of serious disease despite normal reports.

Hoarding → Can't discard items → clutter, poor hygiene.

BDS → Multiple somatic symptoms (pain, fatigue, GI, dizziness) → disproportionate to medical findings.

BODILY DISTRESS SYNDROME

Multiple physical symptoms not fully explained medically

Fatigue, pain, GI symptoms, dizziness, palpitations

Frequent medical visits, somatic complaints, health-seeking

Often believe symptoms are real physical illness

Psychoeducation + CBT (re-attribution therapy)

FMGE 2025,INI CET 23,22

DISSOCIATIVE NEUROLOGICAL SYMPTOM OR CONVERSION SYNDROME

Unconscious production of **neurological symptoms** (motor, sensory, seizures, blindness, paralysis) due to psychological stress.

symptoms are unconscious, patient believes they are real

La belle indifférence (patient unconcerned about serious symptoms)

Patient **not aware** of symptom generation

DISSOCIATIVE AMNESIA

Unconscious circumscribed loss of traumatic memory.

DISSOCIATIVE FUGUE

NEETPG 2025

Sudden, unexpected travel to an unexpected place but a place with emotional significance

May involve confusion about identity, the patient may be confused about his identity

Exam trick:

INICET 20

“Sudden blindness after trauma, but patient unconcerned” → Conversion

“Nurse injects herself with bacteria to get admitted” → Factitious (Munchausen)

“Prisoner fakes chest pain to escape jail” → Malingering

Dissociative Amnesia = Memory lost only

Dissociative Fugue = Flight (travel) + new identity

POST TRAUMATIC STRESS DISORDER

Symptoms : Present ≥ 4 weeks

Hyperarousal/startle response

Nightmares

Flashbacks

Avoidance behaviour

TRT- SSRIs (drug of choice) • CBT (treatment of choice)

Eye Movement Desensitisation & Reprocessing (EMDR) : Most specific for PTSD



ACUTE STRESS DISORDER

NEETPG 2025,FMGE 2024

Similar symptoms to PTSD but time period → 3 days to month

EATING DISORDER

INI CET 2023

SCOFF Questionnaire :

Screening tool for eating disorders. -

S : Feeling sick after eating.

C : Loss of control.

O : Loss of one stone weight (6.35 kg/14 pounds) in 3 months.

F : Erroneous self-perception of being fat.

F : Dominance of food in life.

• Positive : ≥ 2 + Requires further detailed assessment.

Bulimia nervosa	Anorexia nervosa	Binge eating disorder
Recurrent binge eating + compensatory behaviors (vomiting, laxatives, fasting, excessive exercise)	Restriction of food intake → significantly low body weight	Recurrent binge eating without compensatory behaviors
Normal weight/slightly overweight	Underweight (BMI < 18.5, often very low)	overconcern with eating, guilt/shame after binges
Binges followed by purging/fasting	Distorted body image ("I am fat" despite being thin)	Most common eating disorder
Dental caries (enamel erosions) Callouses on knuckles Parotitis (salivary gland inflammations) Hypokalemic and hypochloremic alkalosis	Restrictive eating, excessive exercise, ritualistic behaviors symptoms of starvation such as - amenorrhea, lanugo (neonatal hairs) -hypothermia, dependent edema -bradycardia Endocrine abnormalities (decreased FSH, LH, CRH, increased Cortisol)	-
Treatment- Cognitive behavioural therapy, SSRI	Trt- Hospitalisation is indicated in cases of dehydration, electrolyte imbalance If no weight gain, suspect post-meal purging; supervise for 2 hrs after meals. SSRIs, TCAs, cyproheptadine	-
		

Exam Mnemonics

Anorexia = Absence of weight (thin)

Bulimia = Binge + Barf (purging)

BED = Binge Eating, but no purging

Autism Spectrum Disorder

Persistent deficits in social communication & interaction

Restricted, repetitive patterns of behavior, interests, or activities

Self-destructive behaviours (biting, scratching, head banging)

Onset → before three years of age

Trt- structured classroom teaching

Behavioral therapy

Low-dose antipsychotics (risperidone, aripiprazole) for behavioral issues—not core autism features.

SLEEP DISORDER

EEG RHYTHM	
Alpha wave	Awake, at rest, eyes closed, mind wandering
Beta wave	Awake pattern, when attention is focussed
Theta wave	Transition from wakefulness to sleep, early sleep
Delta wave	Deep sleep

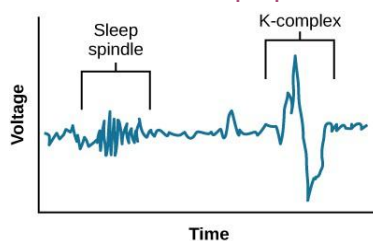
Sleep stages

NEETPG 2020, INICET 22

NREM 1- EEG: Theta waves

NREM 2-Features: Most of sleep time, memory consolidation

EEG: Sleep spindles + K-complexes



NREM 3- EEG: Delta waves (high amplitude, low frequency)

Features: Growth hormone secretion, night terrors, sleepwalking, bedwetting (enuresis)

REM - EEG: Sawtooth waves (like awake state, but paralyzed body)

Increased rate of metabolism in the brain

Penile erection

NIGHT TERROR	NIGHT MARES
NREM (Stage N3 – deep sleep)	REM sleep
No memory of episode	Vivid recall of scary dream
Children (common, benign, outgrow)	Any age
Delta waves (slow-wave sleep)	Saw-tooth waves (REM)

Narcolepsy

↓ Orexin

C/F- Sleep attacks (REM sleep) during daytime.

Cataplexy : Extreme emotional stimuli Hypotonia
Trt- modafanil

SEXUAL DISORDER

Phases of normal sexual response

Desire→ Excitement→ Plateau → Orgasm (shortest phase)→ Resolution/refractory.

NEETPG 2023

Erectile dysfunction	Premature ejaculation
disease of excitement stage Trt- PDE5- inhibitor→ sildenafil,tadalafil	disease of orgasm Ejaculation within a minute of vaginal intercourse Trt- SSRI→ dapoxetine,paroxetine Squeeze technique- master and johnson technique

ATTENTION DEFICIT HYPERACTIVITY DISORDER

NEETPG 2024,FMGE 2024

Onset **before age 12**, symptoms in **≥2 settings** (home + school).

Neurodevelopmental disorder with -

- 1.Inattention- Careless mistakes, easily distracted, forgetful, disorganized.
- 2.Hyperactivity-Fidgety, can't sit still, talks excessively.
- 3.Impulsivity-Interrupts, blurts out answers, can't wait to turn.

Stimulants (first line)

Methylphenidate (blocks DA/NE reuptake) → DOC

Amphetamines (↑ DA/NE release)

Non-stimulants (if stimulant not tolerated)

Atomoxetine (NE reuptake inhibitor)

Exam Tricks

ADHD vs Conduct disorder → ADHD = impulsive but not violating rights of others.

ADHD vs Oppositional Defiant Disorder (ODD) → ODD = argumentative/defiant, but no inattention/hyperactivity.

CONDUCT DISORDER (CD)

FMGE 20

A repetitive, persistent pattern of behavior where the basic rights of others are violated.

May develop into antisocial personality disorder

Features-

Aggression to people & animals → fighting, bullying, cruelty, weapons, sexual assault.

Destruction of property → fire-setting, vandalism.

Progression course-

ODD (Oppositional Defiant Disorder) → Conduct Disorder → Antisocial Personality Disorder (after 18 yrs).

Trt- first line → psychosocial intervention

MENTAL RETARDATION (INTELLECTUAL DISABILITY)

NORMAL	90-109
BORDERLINE RETARDATION	70-89
MILD MENTAL RETARDATION	50-69
MODERATE MENTAL RETARDATION	35-49
SEVERE MENTAL RETARDATION	20-34
PROFOUND MENTAL RETARDATION	<20

Transference- Feeling that the patient develops for the doctor

Countertransference- Feeling the clinician develops for the patient

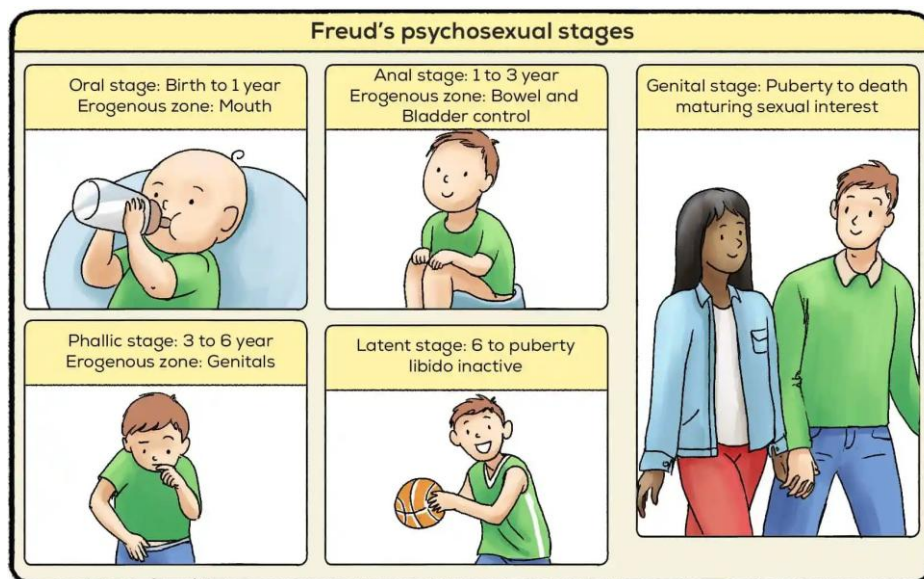
NEETPG 2021

Psychosexual stages of development

FMGE 23

Proposed by Sigmund Freud

1. Oral stage (0-1.5 years)
2. Anal stage (1.5 yrs- 3 yrs)
3. Phallic stage (3yrs-5 yrs)
4. Latent stage (5-12 yrs)
5. Genital stage (12 years till young adulthood)



DEFENSE MECHANISM

FMGE 2024

1. Mature defenses-

A. Suppression → Consciously delaying dealing with stress ("I'll think about exams after my birthday").

B. Anticipation → Planning realistically for future discomfort (studying early for exam)

C. Suppression → Consciously delaying dealing with stress ("I'll think about exams after my birthday").

D. Altruism → Deriving satisfaction by helping others (doctor choosing palliative care).

2. Immature Defenses

- A. Acting out → Expressing feelings through actions, not words (teen slams doors when angry)
- B. Passive-aggression → Indirect resistance (procrastination, intentional inefficiency).
- C. Reverting to earlier childlike behavior under stress ("A 25-year-old man starts sucking his thumb when anxious.")

3. NARCISSISTIC

- A. Denial → Refusing to accept reality (smoker denies risk of cancer).
- B. Projection → Attributing one's own flaws/feelings to others.
- C. Splitting → Seeing others as all good or all bad (common in Borderline Personality Disorder).

ELECTROCONVULSIVE THERAPY

Indirect ECT (Modified ECT)- Anesthetic agents and muscle relaxants are used
Methohexital is the anaesthetic agent of choice
Thiopental and propofol, can also be used
Succinylcholine, used as a muscle relaxant
Atropine, used to decrease secretions

SUBSTANCE RELATED AND ADDICTIVE DISORDERS

Alcohol withdrawal

NEETPG 2024,21,FMGE 2025,24,21,20,INI CET 22

Symptoms that develop after cessation of alcohol

After 6-8 hrs	Tremors - most common
After 12-24 hrs	Alcoholic hallucinations
After 24-48 hrs	Alcohol withdrawal seizures. Usually generalised and tonic-clonic.
After 48-72 hrs	Delirium tremens.

NEETPG 2023,FMGE 2024,23,21,INICET 21, FMGE 2026

Alcohol induced neurocognitive disorder

A. wernicke encephalopathy- acute complication of thiamine deficiency

G-global confusion

O-opthalmoplegia

A-ataxia

Treatment - High dose of parenteral thiamine. Ophthalmoplegia responds first to the treatment

B.Korsakoff syndrome - Chronic complication of thiamine deficiency

Confabulations

Anterograde amnesia

Trt- oral thiamine

DOC for alcohol withdrawal- benzodiazepines

Liver damage then give → lorazepam/oxazepam

Thiamine must be administered before giving glucose to prevent the precipitation of

Wernicke's encephalopathy

For delirium tremens- Intravenous benzodiazepines are the drugs of choice, parenteral thiamine

Screening questionnaire- CAGE questionnaire

C – Cut down:

Have you ever felt you should cut down on your drinking?

A – Annoyed:

Have people annoyed you by criticizing your drinking?

G – Guilty:

Have you ever felt guilty about drinking?

E – Eye-opener:

Have you ever had a drink first thing in the morning (Eye-opener) to steady nerves or get rid of a hangover?

OPIOIDS

M/C abused opioid- heroin

Opioid intoxication

CNS depression (drowsiness → coma)

Respiratory depression (↓ RR, shallow breathing, apnea) → most common cause of death

Miosis (pinpoint pupils) → Exception: severe hypoxia → may cause dilated pupils

DOC- I.V. naloxone (opioid antagonist)

Opioid withdrawal

Dilated pupils (mydriasis)

Lacrimation, rhinorrhea

Piloerection ("cold turkey")

Rhinorrhea, sweating, diarrhoea

Trt- long acting → methadone

Cannabis

FMGE 23

M/C used illegal drug in India and the world

Δ -9-Tetrahydrocannabinol (THC) → main psychoactive substance

Withdrawal	intoxication
Irritability, anxiety ↓ Appetite, insomnia Restlessness, depressed mood	Euphoria, relaxation Dry mouth, tachycardia Amotivational syndrome (chronic use – lack of drive) Conjunctival injection (red eyes)

Flashback phenomenon- Recurrence of cannabis use experience in the absence of current cannabis use.

Amotivational syndrome- May develop in individuals who use cannabis frequently. Characterised by a decrease in the desire to work or compete with others

Running amok- Characterized by a sudden, unprovoked, violent outburst of homicidal behavior, often followed by exhaustion or amnesia.

Trt- benzodiazepines for short term, psychotherapy for long term

Cocaine

NEETPG 2020,INI CET 23,FMGE 20

Block dopamine and NE reuptake transporter

Nasal septal perforation , jet black pigmentation of tongue seen

Intoxication-

c/f- ↑ HR,hypertension,tactile hallucinations or cocaine bugs or magnan phenomenon or formication,sweating,mydriasis

Tobacco

FMGE 22,INICET 21

Most common substance used in India

Nicotine give kick by increasing choline and dopamine

Withdrawal symptoms

Irritability,anxiety,constipation,depression

Trt- nicotine replacement therapy- gums,patches,sprays

Bupropion- 1st line → NDRI- noradrenergic dopamine reuptake inhibitors

Varenicline- most effective α 2 β 4 partial agonism

Delirium- Neurotransmitter involved- Acetylcholine

Most common neurocognitive disorders with an acute onset, common in old age

Acute, transient, fluctuating disturbance of consciousness, attention, orientation, and cognition, usually due to an underlying medical/organic cause.

Orientation defect = earliest sign

Impairment of attention = hallmark

Attention – impaired (most specific sign)

Awareness – reduced, clouded consciousness

Diagnosis- based on clinical symptoms

MMSE - assesses five cognitive functions → max score 30

HEADSSS- Used in adolescent history taking

INI CET 23

Helps screen for risk behaviors (substance use, unsafe sex, suicide)

Mnemonic = HEADSSS → Home, Education, Activities, Drugs, Sexuality, Safety, Suicide

☐ TIMELINE IN PSYCHIATRY

MANIC EPISODE	at least 1 week
HYPOMANIA	At least 4 consecutive days
MAJOR DEPRESSIVE DISORDER	Lasts 6 to 12 months with episodes for 2 or more weeks
MATERNAL BLUES	2-3 days after delivery
BRIEF PSYCHOTIC DISORDER	Last <1 month, usually stress related
SCHIZOPHRENIFORM DISORDER	1 to 6 months
SCHIZOPHRENIA	>6 months
SCHIZOAFFECTIVE DISORDER	Last >2 weeks
ACUTE STRESS DISORDER	Between 3 days and 1 month
PTSD	Lasts >1 month
ADHD	Symptoms .6 months
BULIMIA NERVOSA	At least 3 months

☐

***Stages of sleep**

Awake (eye open)	BETA RHYTHM	
Awake (eye close)	AIPHA RHYTHM	
Stage n1	Light sleep	
Stage n2	Sleep spindles and k complexes	Bruxism
Stage n3	Deep sleep (delta rhythm)	Bed wet , sleep walk ,night terror
REM SLEEP	Paradoxical sleep	Nightmares , saw tooth wave